

Chromatic redshift and Segal conjecture

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Table of Contents

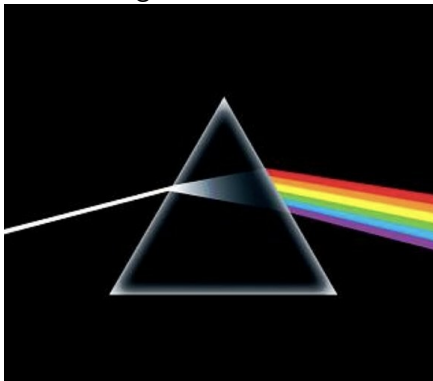
1 Introduction

2 Backgrounds on Algebraic K-theory and Trace method

3 Results and future work



As the title suggests, I should explain the term of 'redshift'. Let's see a picture. As we can see the red-light has **longer** wavelength while the blue-light has **shorter** wavelength.



So, we can make the following definition.

Definition

Redshift is the process by which the wavelength of light is **lengthened**.
Conversely, blueshift is the process by which it is **shortened**.

Back to math, we need to define the wavelength for our studying light (object).

The computations of (stable) homotopy groups of sphere is a central problem. Serre using the Postnikov tower and spectral sequence proved the following result.

Theorem (Serre)

The homotopy groups of sphere spectrum (stable homotopy groups of sphere) $\pi_\mathbb{S}$ are finite abelian groups, for $* > 0$.*

So, it's reasonable to consider the p -primary part of $\pi_*\mathbb{S}$. We know that $\mathrm{Spec}\mathbb{Z} = \{(0), (2), (3), \dots, (p), \dots\}$. The residue fields are $\mathbb{F}_p$. In stable homotopy theory, we have a similar result. The set $\mathrm{Spc}(\mathrm{Sp}^\omega)$ is

$$(2, 0), (2, 1), \dots, (2, n), \dots$$

$$(3, 0), (3, 1), \dots, (3, n), \dots$$

...

$$(p, 0), (p, 1), \dots, (p, n), \dots$$

The corresponding "residue fields" are $K(n, p)$, they are called Morava K-theories.



The Morava K-theory has following properties. The number n is called height.

- 1 For $n = 0$, $K(0, p) = H\mathbb{Q}$.
- 2 For $n > 0$, the homotopy groups of $K(n, p)$ is $\pi_* K(n, p) \simeq \mathbb{F}_p[v_n^{\pm}]$, where $|v_n| = 2(p^n - 1)$.
- 3 $K(1, p)$ is one of summand of KU/p , where KU is the complex K-theory.

Note that $\pi_* KU/p \simeq \mathbb{F}_p[u^{\pm}]$, $|u| = 2$. Hence $u^{p-1} = v_1$ (up to a unit). The Bott periodicity can be viewed as height one periodicity. When considering only Sp_p (the lights), the height (or equivalently the periodicity) can be viewed as the wave length.



In the classical algebra, the base category is the category of abelian groups. This category is a symmetrical monoidal abelian category, the tensor unit is $\mathbb{Z}$. A $\mathbb{Z}$ -algebra is an algebra object in $\mathbf{Ab}$, a commutative algebra is a commutative ring.

In higher algebra, the base category is the category of spectra, denoted by $\mathbf{Sp}$. The objects are stable topological spaces. The category $\mathbf{Sp}$ is a symmetrical monoidal stable ∞ -category. An $\mathbb{S}$ -algebra is an algebra object in $\mathbf{Sp}$, a commutative $\mathbb{S}$ -algebra is an $\mathbb{E}_\infty$ -algebra object in $\mathbf{Sp}$. In fact, there is a natural t -structure on $\mathbf{Sp}$ and

$$\mathbf{Sp}^{\heartsuit} \simeq \mathbf{Ab}.$$

Note that we have a map

$$\mathbb{S} \xrightarrow{p} \mathbb{S}$$

We denote the cofiber of it as $\mathbb{S}/p$, given a spectrum X , we denote $X \otimes \mathbb{S}/p$ by X/p .



Theorem (Hopkins–Smith)

*Let X be a p -local finite spectrum. If $K(n)_*X \simeq 0$ then $K(n-1)_*X \simeq 0$. Moreover, if $K(n)_*X \neq 0$ (type n) then there exists a self map*

$$f : \Sigma^{|v_n^m|} X \rightarrow X$$

*such that $K(n)_*f$ is isomorphism.*

Hence, one can form the spectrum $\mathbb{S}/(p^?, v_1^?, v_2^?, \dots, v_n^?)$ as the iterated cofibers. Similarly, we define $X/(p^?, v_1^?, v_2^?, \dots, v_n^?)$ to be $X \otimes \mathbb{S}/(p^?, v_1^?, v_2^?, \dots, v_n^?)$.

We can use both $K(n)$ and p -local type n spectrum to define v_n -periodicity (wave length).



Remark

Note that the natural map

$$K(1) \simeq K(1)/p \rightarrow v_1^{-1}K(1)/p \simeq v_1^{-1}K(1)$$

is an isomorphism.

Definition

We say a spectrum X has v_n -periodicity if the natural map

$$X/(p^?, v_1^?, v_2^?, \dots, v_{n-1}^?) \rightarrow v_n^{-1}X/(p^?, v_1^?, v_2^?, \dots, v_{n-1}^?)$$

induce isomorphism in π_* when $* \gg 0$. We set $v_0 = p$.



The algebraic K-theory is a lax symmetric monoidal functor

$$K : \text{Alg}(\text{Sp}) \rightarrow \text{Sp}.$$

In fact,

$$K : \text{Cat}_{\infty}^{\text{perf}} \rightarrow \text{Sp}.$$

Roughly speaking, we can ask the following problem.

Chromatic redshift problem

If X is a ring spectrum, and has v_n -periodicity. Does $K(X)$ (or $K(X)_p^{\wedge}$) has v_{n+1} -periodicity.

Example

$\mathbb{Z}[1/p]$ has v_0 -periodicity, $K(\mathbb{Z}[1/p])$ has v_1 -periodicity.

One can also use $K(n)$ to detect v_n -elements.

Theorem (Hahn)

*Let X be an $\mathbb{E}_\infty$ -ring. If $K(n)_*X \simeq 0$ then $K(n+1)_*X \simeq 0$.*

Then one can define height for an $\mathbb{E}_\infty$ -ring. An $\mathbb{E}_\infty$ -ring X is of height n if $K(n)_*X \neq 0$ and $K(n+1)_*X \simeq 0$.

Theorem (Land–Mathew–Meier–Tamme, Yuan, Burklund–Schlank–Yuan)

Let X be an $\mathbb{E}_\infty$ -ring of height n , then $K(X)$ is of height $n+1$.

Example

$\mathbb{F}_p$ is of height -1 , $K(\mathbb{F}_p)$ is of height 0 .

$\pi_*K(\mathbb{F}_p) \otimes \mathbb{Q} \simeq H_*(K_0 \times BGL(\mathbb{F}_p), \mathbb{Q}) \neq 0$. Mitchell proved that $K(1)_*K(\mathbb{Z}) \simeq 0$.

Let R be a ring. Denote $P(R)$ to be set of isomorphism classes of finitely generated projective R -modules.

Definition

Define the 0th algebraic K-theory of R by $K_0(R) := (P(R), \oplus)^{gp}$.

For higher algebraic K-theory, we also use the group completion. Note that the category of finitely generated projective R -modules is a symmetrical monoidal category, taking nerve of this category yields a commutative monoid in $\mathcal{S}$.

There is a group completion functor

$$\mathrm{Alg}_{\mathbb{E}_\infty}(\mathcal{S}) \rightarrow \mathrm{Alg}_{\mathbb{E}_\infty}(\mathcal{S})^{gp} \simeq \mathrm{Sp}_{\geq 0}$$



There is a conjecture made by Lichtenbaum and Quillen.

Conjecture

Let $\mathcal{O}$ be a regular $\mathbb{Z}[1/\ell]$ -algebra, where ℓ is a prime. There is a spectral sequence with E^2 -term

$$E_2^{s,t} = \begin{cases} H_{\text{ét}}^{-s}(\mathcal{O}, \mathbb{Z}/\ell^n(i)), & t = 2i, \\ 0, & t \text{ odd.} \end{cases} \implies K_{s+t}(\mathcal{O}, \mathbb{Z}/\ell^n).$$

converging to the target for $s + t \geq 1$.

This is already a theorem, we briefly sketch the proof.



Theorem (Thomason)

If $\mathcal{O}$ contains an ℓ^n th root of unity, then there is a spectral sequence:

$$E_2^{s,t} = \begin{cases} H_{\text{ét}}^{-s}(\mathcal{O}, \mathbb{Z}/\ell^n(i)), & t = 2i, \\ 0, & t \text{ odd.} \end{cases} \implies K_{s+t}(\mathcal{O}, \mathbb{Z}/\ell^n)[\beta^{-1}].$$

The target can be identified as $L_1^f K(\mathcal{O}) \otimes \mathbb{S}/\ell \simeq K(\mathcal{O}) \otimes \mathbb{S}/\ell[v_1^{-1}]$.
In fact, there is a motivic spectral sequence

$$E_2^{p,q} = H_{\text{mot}}^{p-q}(X, \mathbb{Z}/\ell(-q)) \implies K_{-p-q}(X, \mathbb{Z}/\ell),$$

where X is a smooth scheme over a field k .



Now the conjecture follows from the norm-residue theorem.

Theorem (Voevodsky)

If $\ell \in k^\times$, then there are isomorphisms

$$H_{\text{mot}}^p(X, \mathbb{Z}/\ell(q)) \rightarrow H_{\text{et}}^p(X, \mathbb{Z}/\ell(q))$$

for $p \leq q$.

Conjecture (Waldhausen)

For a nice ring $\mathcal{O}$, the map

$$K(\mathcal{O}) \otimes \mathbb{S}/\ell \rightarrow L_1^f K(\mathcal{O}) \otimes \mathbb{S}/\ell \simeq K(\mathcal{O})/\ell[v_1^{-1}]$$

is an equivalence in large degrees.

Rognes generalized this idea to higher height ring spectra. Fix a prime p .

Definition

- A spectrum X is fp spectrum if X is p -complete, bounded below, and there is a p -local finite spectrum F such that $\pi_*(F \otimes X)$ is finite.
- The spectrum X is said fp type n , if there is a type n finite complex $F \otimes X$ is π -finite, and for any type $n - 1$ finite spectrum W , the spectrum $W \otimes X$ is not π -finite.
- X has pure fp-type n if furthermore F_*X is a free finitely generated $P(v)$ -module for some finite CW spectrum F of type n , with v_n -map $v : \Sigma^d F \rightarrow F$.

Chromatic redshift problem

Let R be an S -algebra of pure fp type n . Does $\mathrm{TC}(R, p)$ have pure fp type $n + 1$?

There is a theorem relating fp type n and v_n -periodicity we defined before.

Theorem (Mahowald–Rezk)

If Y is a spectrum of fp type n , then the localization map

$$Y \rightarrow L_n^f Y$$

is an equivalence on homotopy groups in large degrees. Here L_n^f is the Bousfield localization with respect to $T(0) \oplus \cdots T(n)$.

Remark

Let V be a p -local finite spectrum of type n . Then

$$L_n^f Y \otimes V \simeq v_n^{-1} Y \otimes V.$$

A powerful tool for understanding algebraic K-theory is trace method.

Definition

Let $A \in \text{Alg}(\text{Sp})$. We define

$$\text{THH}(A) := A \otimes_{A \otimes A^{op}} A \in \text{Sp}.$$

Definition

A p -typical cyclotomic spectrum X is a p -complete object $X \in \text{Fun}(BS^1, \text{Sp})$ equipped with an S^1 -equivariant map $\varphi : X \rightarrow X^{tC_p}$.

Definition (Nikolaus–Scholze)

If X is bounded below, p -typical cyclotomic spectrum, then we define

$$\text{TC}(X) := \text{fib}(\varphi^{hS^1} - \text{can} : X^{hS^1} \rightarrow X^{tS^1}).$$

Originally, one uses TR to define TC.

Definition

Let X be a p -typical cyclotomic spectrum. We define

$$\mathrm{TR}^j(X) := X \times_{X^{tC_p}} X^{hC_p} \times_{X^{tC_{p^2}}} X^{hC_{p^2}} \times_{X^{tC_{p^3}}} \cdots \times_{X^{tC_{p^j}}} X^{hC_{p^j}}.$$

$$\mathrm{TR}(X) := \lim_j \mathrm{TR}^j(X).$$

There are maps $F : \mathrm{TR}^n(X) \rightarrow \mathrm{TR}^{n-1}(X)$ corresponding to projecting away from the first factor and then using the inclusion of each C_{p^k} homotopy fixed points into $C_{p^{k-1}}$ homotopy fixed points.

$$\mathrm{TC}(X) := \mathrm{fib}(F - 1 : \mathrm{TR}(X) \rightarrow \mathrm{TR}(X)).$$

Algebraic K-theory is an universal localizing invariant. We have following trace maps

$$K(X) \rightarrow TC(X) \rightarrow TC^-(X) = THH(X)^{hS^1} \rightarrow THH(X).$$

The composition of the maps is called Dennis trace map. The first map is called cyclotomic trace map, it is nil-invariant. Meaning we have the following theorem.

Theorem (Dundas–Goodwillie–McCarthy)

Let $f : R \rightarrow S$ be a map of connective $\mathbb{E}_1$ -algebras. If $\pi_0 f : \pi_0 R \rightarrow \pi_0 S$ has nilpotent kernel, then we have a pull-back square,

$$\begin{array}{ccc} K(R) & \longrightarrow & TC(R) \\ \downarrow & & \downarrow \\ K(S) & \longrightarrow & TC(S). \end{array}$$

Since TC is the fiber of TR, one can ask a stronger question.

Lichtenbaum–Quillen problem

Let X be an S -algebra of pure fp type n . Does $\mathrm{TR}(X)$ has fp type $n + 1$.

Example

- (Hesselholt–Madsen 1997) Let k be a finite field of characteristic p . Then $\mathrm{TR}(k) \simeq HW(k)$.
- (Ausoni–Rognes 2002) The Adams summand ℓ_p is pure fp type 1. $\mathrm{TC}(\ell, p)$ is pure fp type 2.
- (Hahn–Wilson 2022) The truncated Brown–Peterson spectrum $\mathrm{BP}\langle n \rangle_p^\wedge$ is pure fp type n , $\mathrm{TR}(\mathrm{BP}\langle n \rangle_p^\wedge)$ is fp type $n + 1$.



Definition

Let X be a bounded below p -typical cyclotomic spectrum. Let F be a fixed finite spectrum.

- F -Bounded TR for X = The spectrum $F \otimes \mathrm{TR}(X)$ is bounded.
- F -Bounded TC for X = The spectrum $F \otimes \mathrm{TC}(X)$ is bounded.
- F -Segal Conjecture for X = The fiber of the Frobenius $F \otimes X \rightarrow F \otimes X^{tC_p}$ is bounded above.
- F -Weak Canonical Vanishing for X = There is an integer $d \geq 0$, such that for any $i \geq d$ the map

$$\pi_i(F \otimes \mathrm{can}) : \pi_i(F \otimes X^{hC_{p^k}}) \rightarrow \pi_i(F \otimes X^{tC_{p^k}})$$

is zero for all $0 \leq k \leq \infty$.

Theorem (Hahn–Wilson)

Let X be a bounded below, p -typical cyclotomic spectrum and $F \otimes X$ is p -power torsion ($\exists N, p^N = 0 \in \pi_0(\text{Map}(F \otimes X, F \otimes X))$.) Then we have,

$$\boxed{F\text{-Segal Conjecture}} \wedge \boxed{F\text{-Weak Canonical Vanishing}} \iff \boxed{F\text{-Bounded TR}}$$

Definition

A $\mathbb{E}_n$ -MU-algebra B is called form of $\text{BP}\langle n \rangle$, if

$$\mathbb{Z}_{(p)}[v_1, v_2, \dots, v_n] \subseteq \text{BP}_* \subseteq \pi_* \text{MU}_{(p)} \rightarrow \pi_* B$$

is isomorphism.

By Lawson's work, there is no $\mathbb{E}_{12}$ -MU-algebra form of $\text{BP}\langle n \rangle$.



Remark

- Hahn–Wilson showed that there is an $\mathbb{E}_3$ -MU-algebra form of $BP\langle n \rangle$.
- Let F be a type $n + 2$ complex. Hahn–Wilson proved that F -Bounded TR is true for $THH(BP\langle n \rangle)$.

One can further proves the following.

Theorem (Hahn–Wilson, Mathew)

Suppose A is a connective $\mathbb{E}_1$ - $BP\langle n \rangle$ -algebra and that F is type $n + 2$ complex. Then if F -Segal conjecture is true for $THH(A)$, $F \otimes TR(A)$ is bounded.



Construction

- We define $\mathbb{S}[x] := \Sigma_+^\infty \mathbb{N}$.
- For $e \in \mathbb{N}_{\geq 1}$, $M_e := \{1, x, \dots, x^{e-1}\} \simeq \mathbb{N}/[e, \infty)$ where the multiplication is defined in an obvious way. Let $\mathbb{S}[x]/x^e := \Sigma_+^\infty M_e \in \mathcal{S}\mathcal{p}$.
- The multiplication of $M_e, \mathbb{N}$ are strictly commutative. We learn that $\mathbb{N}, M_e$ are commutative monoid objects in $\mathcal{S}$, and hence commutative algebra objects in $\mathcal{S}$. So, $\mathbb{S}[x], \mathbb{S}[x]/x^e \in \mathcal{C}\mathcal{A}\mathcal{l}\mathcal{g}$.
- We define $\mathcal{BP}\langle n \rangle[x] := \mathcal{BP}\langle n \rangle \otimes \mathbb{S}[x]$. We define $\mathcal{BP}\langle n \rangle[x]/x^e := \mathcal{BP}\langle n \rangle \otimes \mathbb{S}[x]/x^e$. Since $\text{Mod}_{\mathbb{S}} \rightarrow \text{Mod}_{\mathbb{S}[x]}$ and $\text{Mod}_{\mathbb{S}} \rightarrow \text{Mod}_{\mathbb{S}[x]/x^e}$ are symmetric monoidal, so those algebras are $\mathbb{E}_3$.

Theorem (Fang)

Let F be any type $n + 1$ finite complex, then the Frobenius

$$F_*(\mathrm{THH}(\mathrm{BP}\langle n \rangle[x])) \rightarrow F_*(\mathrm{THH}(\mathrm{BP}\langle n \rangle[x])^{tC_p})$$

induces isomorphism on large degrees. In particular, if F is of type $n + 2$, the above also holds.

Theorem (Fang)

Let F be any p -local finite type 3 complex. Then the following spectra are bounded.

$$\begin{aligned} F \otimes \mathrm{TC}(\mathrm{BP}\langle 1 \rangle[x]), \\ F \otimes \mathrm{K}(\mathrm{BP}\langle 1 \rangle[x])_p^\wedge. \end{aligned}$$

Corollary

Consequently, the following maps have bounded above fibers

$$\begin{aligned} \mathrm{TC}(\mathrm{BP}\langle 1\rangle[x]) \otimes \mathbb{S}/p &\rightarrow L_2^f(\mathrm{TC}(\mathrm{BP}\langle 1\rangle[x]) \otimes \mathbb{S}/p), \\ \mathrm{K}(\mathrm{BP}\langle 1\rangle[x]) \otimes \mathbb{S}/p &\rightarrow L_2^f(\mathrm{K}(\mathrm{BP}\langle 1\rangle[x]) \otimes \mathbb{S}/p). \end{aligned}$$

Theorem (Fang)

Let F be any type $n + 2$ finite complex. Then the fiber of the Frobenius

$$F \otimes \mathrm{THH}(\mathrm{BP}\langle n\rangle[x]/x^e) \rightarrow F \otimes \mathrm{THH}(\mathrm{BP}\langle n\rangle[x]/x^e)^{tC_p}$$

is not bounded above, where $e > 1$. Consequently, $F \otimes \mathrm{TR}(\mathrm{BP}\langle n\rangle[x]/x^e)$ is not bounded, where $e > 1$.

Sketch of proof

- We apply the descent functor filtering the Frobenius map as

$$\mathrm{THH}(\mathrm{desc}_{\mathbb{F}_p}(\mathrm{BP}\langle n\rangle[x]))_{v_0}^{\wedge} \otimes_{\mathrm{desc}_{\mathbb{F}_p} \mathrm{BP}\langle n\rangle} \mathbb{F}_p \rightarrow$$

$$\mathrm{THH}(\mathrm{desc}_{\mathbb{F}_p} \mathrm{BP}\langle n\rangle[x])^{tC_p} \otimes_{\mathrm{desc}_{\mathbb{F}_p}(\mathrm{BP}\langle n\rangle)} \mathbb{F}_p$$
- The construction $\mathrm{desc}_{\mathbb{F}_p} : \mathrm{Sp} \rightarrow \mathrm{Fil}(\mathrm{Mod}_{\mathbb{F}_p})$ is lax symmetric monoidal.
- Note that the associated graded has homotopy groups as

$$\pi_*(\mathrm{gr}_{\bullet}(\mathrm{BP}\langle n\rangle[x])) \simeq \mathrm{Ext}_{\mathcal{A}_*}^{*,*}(\mathbb{F}_p, H_*(\mathrm{BP}\langle n\rangle[x], \mathbb{F}_p)).$$

- $\pi_* \mathrm{gr}_{\bullet}(\mathrm{BP}\langle n\rangle[x]) \simeq \mathrm{Ext}_{\Lambda(\overline{\tau}_0, \dots, \overline{\tau}_n)}(\mathbb{F}_p, \mathbb{F}_p[x]) \simeq \mathbb{F}_p[x, v_0, v_1, \dots, v_n]$
- A standard argument of Kozul duality for graded $\mathbb{E}_2$ -algebra shows that $\mathrm{gr}_{\bullet}(\mathrm{BP}\langle n\rangle[x]) \simeq \mathbb{F}_p[x] \otimes \mathbb{S}[a_0] \otimes \mathbb{S}[a_1] \otimes \cdots \mathbb{S}[a_n] = R$.



Sketch of proof

- A standard argument showing that the map

$$\pi_*(\mathrm{THH}(R))/(a_1, a_2, \dots, a_n) \rightarrow \pi_*(\mathrm{THH}(R)^{tC_p})/(a_1, a_2, \dots, a_n)$$

is an equivalence in degrees $* > n + 1 + \sum_{i=1}^n |a_i|$.

- Note that the thick subcategory of type $n + 1$ finite spectra is generated by the generalized Moore spectra $\mathbb{S}/(p^{i_0}, v_1^{i_1}, \dots, v_n^{i_n})$. So for any type $n + 1$ finite spectrum F , one can show the F -Segal conjecture is true for $\mathrm{THH}(\mathrm{BP}\langle n \rangle[x])$.

Remark

These arguments still valid for $\mathrm{THH}(\mathrm{BP}\langle n \rangle[x_1, x_2, \dots, x_m])$.

Remark

- The base case is $\mathrm{THH}(\mathbb{F}_p[x])$. L. Hesselholt shows that the $\mathbb{S}/p$ -Segal conjecture is true for $\mathrm{THH}(k[x])$, where k is a perfect field of characteristic p .
- There is a construction $\mathbb{S}_W : \mathrm{CAlg}_{\mathbb{F}_p}^{\mathrm{perf}} \rightarrow \mathrm{CAlg}_{\mathbb{S}_p}$, such that $\mathbb{S}_{W(k)} \otimes_{\mathbb{S}_p} \mathbb{F}_p \simeq k$.
- So, we have $H_*(\mathrm{BP}\langle n \rangle \otimes \mathbb{S}_{W(k)}) \simeq H_*(\mathrm{BP}\langle n \rangle) \otimes_{\mathbb{F}_p} H_*(\mathbb{S}_{W(k)}) \simeq H_*(\mathrm{BP}\langle n \rangle) \otimes_{\mathbb{F}_p} k$.
- Hence, one can also prove the F -Segal conjecture for $\mathrm{THH}(\mathrm{BP}\langle n \rangle \otimes \mathbb{S}_{W(k)}[x])$, where F is a p -local type $n+2$ finite spectrum.



Sketch of proof

Theorem (Hesselholt–Madsen)

Let $V_d = \mathbb{C}(\xi_m) \oplus \cdots \oplus \mathbb{C}(\xi_m^d)$, where ξ_m is the m th primitive root of unity. We use S^{V_d} to denote the one-point compactification of V_d . Then $\mathrm{THH}(\mathbb{S}[x]/x^e) \simeq \bigoplus_{m \geq 0} B_m$, as S^1 -spectra, where

$$B_m \simeq S^{V_{\lfloor \frac{m-1}{e} \rfloor}} \otimes \Sigma_+^\infty S^1 / C_m,$$

if $e \nmid m$, and

$$B_m \simeq S^{V_{\lfloor \frac{m-1}{e} \rfloor}} \otimes \mathrm{cofib}(\Sigma_+^\infty S^1 / C_{m/e} \rightarrow \Sigma_+^\infty S^1 / C_m)$$

if $e \mid m$.

Sketch of proof

- By formal argument, we need to understand the unstable Frobenius $F \otimes \mathrm{THH}(\mathrm{BP}\langle n \rangle) \otimes B_m \rightarrow F \otimes \mathrm{THH}(\mathrm{BP}\langle n \rangle)^{tC_p} \otimes B_m \simeq F \otimes (\mathrm{THH}(\mathrm{BP}\langle n \rangle) \otimes B_{mp})^{tC_p}$.
- One sees that B_m increasing the connectivity of the fiber of Frobenius map.
- Using the descent functor $\mathrm{desc}_{\mathbb{F}_p}$ and argument presented before, one can shows that the fiber of Frobenius map

$$F \otimes \mathrm{THH}(\mathrm{BP}\langle n \rangle) \otimes B_m \rightarrow F \otimes \mathrm{THH}(\mathrm{BP}\langle n \rangle)^{tC_p} \otimes B_m$$

is nonzero.

- Therefore, the fiber of the total Frobenius map is not bounded above

Remark

- In fact, one can use the decomposition of $\mathrm{THH}(\mathbb{S}[x]/x^e) \simeq \bigoplus_{m \geq 0} B_m$ to partially compute $V(2)_* \mathrm{TC}(\ell[x]/x^e)$.
- Basically by the decomposition, and calculations done by Asouni–Rognes, one can show that $V(2)_* \mathrm{TC}(\ell[x]/x^e)$ is not bounded.
- However, $\mathrm{THH}(\mathrm{BP}[x]/x^e)$ satisfies $\mathbb{S}/p$ -Segal conjecture. Because the fiber

$$\mathbb{S}/p \otimes \mathrm{THH}(\mathrm{BP}) \rightarrow \mathbb{S}/p \otimes \mathrm{THH}(\mathrm{BP})^{tC_p}$$

is zero.

- So, in some sense, the regularity is essential.

